I Hate Mathematics

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Hi, my name is Vectra, I am a normal university student at the University of Tears, Anxiety, Deadlines, Exams and Overthinking (UTADEO) who really really hates math. My degree is systems engineering but all math doesn't matter there, I don't think that this bunch of math is ever going to be useful in life.

Me and my friends love to *parchar* in “La Brújula”. The other day, a random guy came to us shouting a bunch of mathematical formulas that I didn't understand. In his madness he said that all of those formulas lead to the secret of the universe.

Although I don't like math, I do feel curious about the universe... can you help me calculate the numeric results of the formulas that he said?

You are given several queries. Each query corresponds to a specific mathematical formula. Your task is to evaluate each query and print its result.

- **ADD**$(a, b) = a + b$
- **MUL**$(a, b) = a \times b$
- **POW**$(a, b) = a^b$ (using a loop, NOT built-in power)
- **FACT**$(n) = n!$ (using a recursive function)
- **POLY**$(a, b) = a^2 + 2ab + b^2$
- **DIST**$(x_1, y_1, x_2, y_2) = (x_1 - x_2)^2 + (y_1 - y_2)^2$
- **AVG**$(a, b, c) = \frac{a+b+c}{3}$
- **SECRET OF UNIVERSE**$(n, a, b) =$

$$\sum_{i=1}^{n} \left(\sum_{j=1}^{n} \left(\prod_{k=1}^{n} \left(\pi \times e^{i+j+k} + (a^2 + 2ab + b^2) + \frac{a+b+n}{3} \right) \right) \right)$$

Program input

The first line contains an integer q $(1 \leq q \leq 100)$, the number of queries.

Each of the next q lines contains one of the following queries:

- `ADD a b`
- `MUL a b`
- `POW a b`
- `FACT n`
- `POLY a b`
- `DIST x1 y1 x2 y2`
- `AVG a b c`

- `UNIVERSE n a b`

All values are integers in the range:

- General values: $[-10^3, 10^3]$
- Factorial: $0 \leq n \leq 10$
- SECRET OF UNIVERSE: $1 \leq n, a, b \leq 30$

Program output

For each query, print a single line with the result. Any decimal value must be rounded to two decimal places.

Sample test cases

Prueba #1

Input	Output

`8`
`ADD 1 1`
`MUL 2 2`
`POW 3 3`
`POLY 2 3`
`DIST 0 0 3 4`
`AVG 3 4 8`
`FACT 5`
`UNIVERSE 2 2 1`
	`2`
	`4`
	`27`
	`25`
	`1`
	`5`
	`120`
	`796762.03`

Notes

- You must build complex functions using simpler ones.
- Avoid using built-in shortcuts such as `**`, `math.pow()`, or `math.factorial()`.
- You can use constant such as `math.e` for Euler number (e) and `math.pi` for pi π.

Important aspects

1. The exam will be completed in **2 stages**:
   - Algorithm development on paper **(3.0 points)**.
     - You must write and justify the time and memory complexity of your algorithm.
     - You must identify the complexity of each function and the overall solution.
   - Implementation in the code editor **(2.0 points)**.
2. The data on each query line is provided as a string. You may use:

```
data = input().split()
```

3. This exercise evaluates the topics covered up to **functions and loops**.
4. You must avoid lambda functions.
5. It is not necessary to use advanced data structures or external libraries.
6. Output format must match exactly.